Lab2 Dimentionality reduction

Source: MLL/Lab2 Dimentionality reduction.ipynb

CELL 1 — CODE | Execution: 1

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

CELL 2 — CODE | Execution: 3

```
df=pd.read_csv("Wine.csv")
```

CELL 3 — CODE | Execution: 3

```
df
```

OUTPUT

```
     Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0      14.23        1.71  2.43          15.6        127           2.80
1      13.20        1.78  2.14          11.2        100           2.65
2      13.16        2.36  2.67          18.6        101           2.80
3      14.37        1.95  2.50          16.8        113           3.85
4      13.24        2.59  2.87          21.0        118           2.80
..       ...         ...   ...           ...        ...            ...
173    13.71        5.65  2.45          20.5         95           1.68
174    13.40        3.91  2.48          23.0        102           1.80
175    13.27        4.28  2.26          20.0        120           1.59
176    13.17        2.59  2.37          20.0        120           1.65
177    14.13        4.10  2.74          24.5         96           2.05

     Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue  \
0          3.06                  0.28             2.29             5.64  1.04
1          2.76                  0.26             1.28             4.38  1.05
2          3.24                  0.30             2.81             5.68  1.03
3          3.49                  0.24             2.18             7.80  0.86
4          2.69                  0.39             1.82             4.32  1.04
..          ...                   ...              ...              ...   ...
173        0.61                  0.52             1.06             7.70  0.64
174        0.75                  0.43             1.41             7.30  0.70
175        0.69                  0.43             1.35            10.20  0.59
176        0.68                  0.53             1.46             9.30  0.60
177        0.76                  0.56             1.35             9.20  0.61

     OD280  Proline  Customer_Segment
0     3.92     1065                 1
1     3.40     1050                 1
2     3.17     1185                 1
3     3.45     1480                 1
4     2.93      735                 1
..     ...      ...               ...
173   1.74      740                 3
174   1.56      750                 3
175   1.56      835                 3
176   1.62      840                 3
177   1.60      560                 3

[178 rows x 14 columns]
```

CELL 4 — CODE | Execution: 5

```
df.info()
```

OUTPUT

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 178 entries, 0 to 177
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Alcohol               178 non-null    float64
 1   Malic_Acid            178 non-null    float64
 2   Ash                   178 non-null    float64
 3   Ash_Alcanity          178 non-null    float64
 4   Magnesium             178 non-null    int64  
 5   Total_Phenols         178 non-null    float64
 6   Flavanoids            178 non-null    float64
 7   Nonflavanoid_Phenols  178 non-null    float64
 8   Proanthocyanins       178 non-null    float64
 9   Color_Intensity       178 non-null    float64
 10  Hue                   178 non-null    float64
 11  OD280                 178 non-null    float64
 12  Proline               178 non-null    int64  
 13  Customer_Segment      178 non-null    int64  
dtypes: float64(11), int64(3)
memory usage: 19.6 KB
```

CELL 5 — CODE | Execution: 6

```
df.describe()
```

OUTPUT

```
          Alcohol  Malic_Acid         Ash  Ash_Alcanity   Magnesium  \
count  178.000000  178.000000  178.000000    178.000000  178.000000   
mean    13.000618    2.336348    2.366517     19.494944   99.741573   
std      0.811827    1.117146    0.274344      3.339564   14.282484   
min     11.030000    0.740000    1.360000     10.600000   70.000000   
25%     12.362500    1.602500    2.210000     17.200000   88.000000   
50%     13.050000    1.865000    2.360000     19.500000   98.000000   
75%     13.677500    3.082500    2.557500     21.500000  107.000000   
max     14.830000    5.800000    3.230000     30.000000  162.000000   

       Total_Phenols  Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  \
count     178.000000  178.000000            178.000000       178.000000   
mean        2.295112    2.029270              0.361854         1.590899   
std         0.625851    0.998859              0.124453         0.572359   
min         0.980000    0.340000              0.130000         0.410000   
25%         1.742500    1.205000              0.270000         1.250000   
50%         2.355000    2.135000              0.340000         1.555000   
75%         2.800000    2.875000              0.437500         1.950000   
max         3.880000    5.080000              0.660000         3.580000   

       Color_Intensity         Hue       OD280      Proline  Customer_Segment  
count       178.000000  178.000000  178.000000   178.000000        178.000000  
mean          5.058090    0.957449    2.611685   746.893258          1.938202  
std           2.318286    0.228572    0.709990   314.907474          0.775035  
min           1.280000    0.480000    1.270000   278.000000          1.000000  
25%           3.220000    0.782500    1.937500   500.500000          1.000000  
50%           4.690000    0.965000    2.780000   673.500000          2.000000  
75%           6.200000    1.120000    3.170000   985.000000          3.000000  
max          13.000000    1.710000    4.000000  1680.000000          3.000000  
```

CELL 6 — CODE | Execution: 7

```
df.head()
```

OUTPUT

```
   Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0    14.23        1.71  2.43          15.6        127           2.80   
1    13.20        1.78  2.14          11.2        100           2.65   
2    13.16        2.36  2.67          18.6        101           2.80   
3    14.37        1.95  2.50          16.8        113           3.85   
4    13.24        2.59  2.87          21.0        118           2.80   

   Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue  \
0        3.06                  0.28             2.29             5.64  1.04   
1        2.76                  0.26             1.28             4.38  1.05   
2        3.24                  0.30             2.81             5.68  1.03   
3        3.49                  0.24             2.18             7.80  0.86   
4        2.69                  0.39             1.82             4.32  1.04   

   OD280  Proline  Customer_Segment  
0   3.92     1065                 1  
1   3.40     1050                 1  
2   3.17     1185                 1  
3   3.45     1480                 1  
4   2.93      735                 1  
```

CELL 7 — CODE | Execution: 8

```
df.tail()
```

OUTPUT

```
     Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
173    13.71        5.65  2.45          20.5         95           1.68
174    13.40        3.91  2.48          23.0        102           1.80
175    13.27        4.28  2.26          20.0        120           1.59
176    13.17        2.59  2.37          20.0        120           1.65
177    14.13        4.10  2.74          24.5         96           2.05

     Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue  \
173        0.61                  0.52             1.06              7.7  0.64
174        0.75                  0.43             1.41              7.3  0.70
175        0.69                  0.43             1.35             10.2  0.59
176        0.68                  0.53             1.46              9.3  0.60
177        0.76                  0.56             1.35              9.2  0.61

     OD280  Proline  Customer_Segment
173   1.74      740                 3
174   1.56      750                 3
175   1.56      835                 3
176   1.62      840                 3
177   1.60      560                 3
```

CELL 8 — CODE | Execution: 9

```
df.dtypes
```

OUTPUT

```
Alcohol                 float64
Malic_Acid              float64
Ash                     float64
Ash_Alcanity            float64
Magnesium                 int64
Total_Phenols           float64
Flavanoids              float64
Nonflavanoid_Phenols    float64
Proanthocyanins         float64
Color_Intensity         float64
Hue                     float64
OD280                   float64
Proline                   int64
Customer_Segment          int64
dtype: object
```

CELL 9 — CODE | Execution: 10

```
df.shape
```

OUTPUT

```
(178, 14)
```

CELL 10 — CODE | Execution: 12

```
df.isnull()
```

OUTPUT

```
      Alcohol  Malic_Acid    Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0       False       False  False         False      False          False
1       False       False  False         False      False          False
2       False       False  False         False      False          False
3       False       False  False         False      False          False
4       False       False  False         False      False          False
..        ...         ...    ...           ...        ...            ...
173     False       False  False         False      False          False
174     False       False  False         False      False          False
175     False       False  False         False      False          False
176     False       False  False         False      False          False
177     False       False  False         False      False          False

      Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity  \
0          False                 False            False            False
1          False                 False            False            False
2          False                 False            False            False
3          False                 False            False            False
4          False                 False            False            False
..           ...                   ...              ...              ...
173        False                 False            False            False
174        False                 False            False            False
175        False                 False            False            False
176        False                 False            False            False
177        False                 False            False            False

        Hue  OD280  Proline  Customer_Segment
0     False  False    False             False
1     False  False    False             False
2     False  False    False             False
3     False  False    False             False
4     False  False    False             False
..      ...    ...      ...               ...
173   False  False    False             False
174   False  False    False             False
175   False  False    False             False
176   False  False    False             False
177   False  False    False             False

[178 rows x 14 columns]
```

CELL 11 — CODE | Execution: 13

```
df.notnull()
```

OUTPUT

```
     Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0       True        True  True          True       True           True
1       True        True  True          True       True           True
2       True        True  True          True       True           True
3       True        True  True          True       True           True
4       True        True  True          True       True           True
..       ...         ...   ...           ...        ...            ...
173     True        True  True          True       True           True
174     True        True  True          True       True           True
175     True        True  True          True       True           True
176     True        True  True          True       True           True
177     True        True  True          True       True           True

     Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue  \
0          True                  True             True             True  True
1          True                  True             True             True  True
2          True                  True             True             True  True
3          True                  True             True             True  True
4          True                  True             True             True  True
..          ...                   ...              ...              ...   ...
173        True                  True             True             True  True
174        True                  True             True             True  True
175        True                  True             True             True  True
176        True                  True             True             True  True
177        True                  True             True             True  True

     OD280  Proline  Customer_Segment
0     True     True              True
1     True     True              True
2     True     True              True
3     True     True              True
4     True     True              True
..     ...      ...               ...
173   True     True              True
174   True     True              True
175   True     True              True
176   True     True              True
177   True     True              True

[178 rows x 14 columns]
```

CELL 12 — CODE | Execution: 14

```
df.isnull().sum()
```

OUTPUT

```
Alcohol                0
Malic_Acid             0
Ash                    0
Ash_Alcanity           0
Magnesium              0
Total_Phenols          0
Flavanoids             0
Nonflavanoid_Phenols   0
Proanthocyanins        0
Color_Intensity        0
Hue                    0
OD280                  0
Proline                0
Customer_Segment       0
dtype: int64
```

CELL 13 — CODE | Execution: 17

```
df.columns
```

OUTPUT

```
Index(['Alcohol', 'Malic_Acid', 'Ash', 'Ash_Alcanity', 'Magnesium',
       'Total_Phenols', 'Flavanoids', 'Nonflavanoid_Phenols',
       'Proanthocyanins', 'Color_Intensity', 'Hue', 'OD280', 'Proline',
       'Customer_Segment'],
      dtype='object')
```

CELL 14 — CODE | Execution: 4

```
X = df.drop("Customer_Segment", axis=1)
y = df["Customer_Segment"]
print("\nNumber of Input Features:",
X.shape[1])
print("\nNumber of Classes:",
y.nunique())
print("\nClasses:")
print(y.unique())
```

OUTPUT

```
Number of Input Features: 13

Number of Classes: 3

Classes:
[1 2 3]
```

CELL 15 — CODE | Execution: 6

```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_state=42,stratify=y)
```

CELL 16 — CODE | Execution: 7

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

CELL 17 — CODE | Execution: 11

```
print("MODEL USING ORIGINAL FEATURES")

model_original=LogisticRegression(max_iter=1000)
model_original.fit(X_train_scaled, y_train)
y_pred_original=model_original.predict(X_test_scaled)
accuracy_original=accuracy_score(y_test,y_pred_original)

print("\nNumber of Original Features:",X_train_scaled.shape[1])
print("Accuracy:",accuracy_original)
```

OUTPUT

```
MODEL USING ORIGINAL FEATURES

Number of Original Features: 13
Accuracy: 0.9722222222222222
```

CELL 18 — CODE | Execution: 12

```
print("DIMENSIONALITY REDUCTION USING PCA")

pca = PCA(n_components=2)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
```

OUTPUT

```
DIMENSIONALITY REDUCTION USING PCA
```

CELL 19 — CODE | Execution: 13

```
print("\nOriginal Number of Features:",X_train_scaled.shape[1])
print("Number of PCA Components:",X_train_pca.shape[1])
print("\nVariance Explained by Each Component:")
print(pca.explained_variance_ratio_)
print("\nTotal Variance Explained:")
print(sum(pca.explained_variance_ratio_))
```

OUTPUT

```
Original Number of Features: 13
Number of PCA Components: 2

Variance Explained by Each Component:
[0.35792104 0.19270671]

Total Variance Explained:
0.5506277578733126
```

CELL 20 — CODE | Execution: 14

```
model_pca = LogisticRegression(max_iter=1000)
model_pca.fit(X_train_pca,y_train)
y_pred_pca = model_pca.predict(X_test_pca)
accuracy_pca = accuracy_score(y_test,y_pred_pca)

print("\nAccuracy using PCA:",accuracy_pca)
```

OUTPUT

```
Accuracy using PCA: 0.9166666666666666
```

CELL 21 — CODE | Execution: 15

```
lda = LinearDiscriminantAnalysis(n_components=2)
X_train_lda = lda.fit_transform(X_train_scaled,y_train)
X_test_lda = lda.transform(X_test_scaled)

print("\nOriginal Number of Features:",X_train_scaled.shape[1])
print("Number of LDA Components:",X_train_lda.shape[1])
```

OUTPUT

```
Original Number of Features: 13
Number of LDA Components: 2
```

CELL 22 — CODE | Execution: 16

```
model_lda = LogisticRegression(max_iter=1000)
model_lda.fit(X_train_lda,y_train)
y_pred_lda = model_lda.predict(X_test_lda)
accuracy_lda = accuracy_score(y_test,y_pred_lda)

print("\nAccuracy using LDA:",accuracy_lda)
```

OUTPUT

```
Accuracy using LDA: 0.9722222222222222
```

CELL 23 — CODE | Execution: 17

```
results= pd.DataFrame({
    "Method":[
        "Original_Feature",
        "PCA",
        "LDA"
    ],
    "Number of Features": [
        X_train_scaled.shape[1],
        X_train_pca.shape[1],
        X_train_lda.shape[1]
    ],
    "Accuracy": [
        accuracy_original,
        accuracy_pca,
        accuracy_lda
    ]
})
print(results)
```

OUTPUT

```
              Method  Number of Features  Accuracy
0  Original_Feature                  13  0.972222
1               PCA                   2  0.916667
2               LDA                   2  0.972222
```

CELL 24 — CODE | Execution: 20

```
plt.figure(figsize=(8, 6))
plt.scatter(
    X_test_pca[:, 0],
    X_test_pca[:, 1],
    c=y_test,
    alpha=0.7
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Wine Dataset after PCA")
plt.colorbar(label="Customer_Segment")

plt.show()
```

OUTPUT

```
<Figure size 800x600 with 2 Axes>
```

CELL 25 — CODE | Execution: 21

```
plt.figure(figsize=(8, 6))
plt.scatter(
    X_test_lda[:, 0],
    X_test_lda[:, 1],
    c=y_test,
    alpha=0.7
)
plt.xlabel("Linear Descriminant 1")
plt.ylabel("Linear Descriminant 2")
plt.title("Wine Dataset after LDA")
plt.colorbar(label="Customer_Segment")

plt.show()
```

OUTPUT

```
<Figure size 800x600 with 2 Axes>
```

CELL 26 — CODE | Execution: 22

```
print(
    classification_report(
        y_test,
        y_pred_original
    )
)
print(
    classification_report(
        y_test,
        y_pred_pca
    )
)
print(
    classification_report(
        y_test,
        y_pred_lda
    )
)
```

OUTPUT

```
              precision    recall  f1-score   support

           1       1.00      1.00      1.00        12
           2       0.93      1.00      0.97        14
           3       1.00      0.90      0.95        10

    accuracy                           0.97        36
   macro avg       0.98      0.97      0.97        36
weighted avg       0.97      0.97      0.97        36

              precision    recall  f1-score   support

           1       0.92      0.92      0.92        12
           2       0.87      0.93      0.90        14
           3       1.00      0.90      0.95        10

    accuracy                           0.92        36
   macro avg       0.93      0.92      0.92        36
weighted avg       0.92      0.92      0.92        36

              precision    recall  f1-score   support

           1       1.00      1.00      1.00        12
           2       0.93      1.00      0.97        14
           3       1.00      0.90      0.95        10

    accuracy                           0.97        36
   macro avg       0.98      0.97      0.97        36
weighted avg       0.97      0.97      0.97        36
```

CELL 27 — CODE

(Empty cell)